Unit: 6 Unit 8: Quadratics **Name:** ____________

Topic: Definition of Quadratics and Standard Form **Date:** ________

Advanced (Habanero)

Q1. What is the standard form of a quadratic function?

(A) $ax^2 + bx + c$

(B) $ax + bx^2 + c$

(C) $ax^2 - bx + c$

(D) $ax^2 + b$

(E) $ax + b$

Q2. Which equation is a quadratic function in standard form?

(A) $x^2 + 3x - 2$

(B) $x^2 - 3x$

(C) $x + 2$

(D) $x^3 + 2x$

(E) $x^2 - 3x + 2x^2$

Q3. What is the value of a in the quadratic function $2x^2+3x-1$?

(A) 1

(B) 2

(C) 3

(D) -1

(E) -2

Q4. Which of the following is a key feature of the quadratic function $x^2 + 2x + 1$?

(A) Vertex at $(0, 0)$

(B) Vertex at $(1, 0)$

(C) Vertex at $(-1, 0)$

(D) Vertex at $(-1, 1)$

(E) No vertex

Q5. What is the axis of symmetry of the quadratic function $x^2 - 4x + 4$?

(A) $x = -2$

(B) $x = -1$

(C) $x = 0$

(D) $x = 1$

(E) $x = 2$

Q6. Which quadratic function has a vertex at $(0, 0)$?

(A) $x^2 + 1$

(B) $x^2 - 1$

(C) x^2

(D) $x + 1$

(E) $x - 1$

Q7. What is the y-intercept of the quadratic function x^2+2x+1?

(A) $(-2, 0)$

(B) $(-1, 0)$

(C) $(-1, 1)$

(D) $(0, 1)$

(E) $(0, 0)$

Q8. Which of the following quadratic functions has no real roots?

(A) $x^2 + 1$

(B) $x^2 - 1$

(C) $x^2 + 2x + 1$

(D) $x^2 - 2x + 1$

(E) x^2

Open Ended Questions (FRQ)

Q9. Graph the quadratic function $f(x) = x^2 - 2x - 3$. Identify the vertex, axis of symmetry, and y-intercept.

Q10. Find the quadratic function in standard form with vertex $(2, -3)$ and y-intercept $(0, 1)$. Show all work and explain your reasoning.

Chef's Tips

- Emphasize the importance of identifying key features of quadratic functions
- Encourage students to use graphing calculators to visualize quadratic functions
- Remind students to check their work and explain their reasoning